



AI-Based Ecological Predictive Cruise Control for Connected EVs

Does an LSTM traffic-speed forecast make an EV's cruise control more energy-efficient? An evaluation on the EPA city, highway and aggressive drive cycles.

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Abstract

A stacked LSTM forecasts the traffic speed 2 seconds ahead. Two predictive controllers use that forecast: a PID with deceleration feedforward (LSTM-PID) and a receding-horizon model predictive controller (LSTM-MPC). They are compared with bang-bang and plain PID control on a 2,500 kg EV model that follows the EPA UDDS, HWFET and US06 drive cycles.

Every forecast is bracketed: persistence (no forecast), constant-acceleration extrapolation, the LSTM, and an oracle that knows the true future. Energy is compared at equal tracking error, because any controller can save energy by following the reference more loosely.

- ❏ A perfect 2 s preview saves **0.7–4.3% energy** at the PID's tracking error. The LSTM recovers little of that: +1.0%, 0.0% and -0.9% on the city, highway and aggressive cycles (5 seeds).



Introduction

The Idea

Cruise control that reacts to speed error brakes late and accelerates hard. A connected EV that knows a slowdown is coming can coast early instead, and the LSTM + MPC design of Zhou et al. (2021) does exactly that.

The Open Question

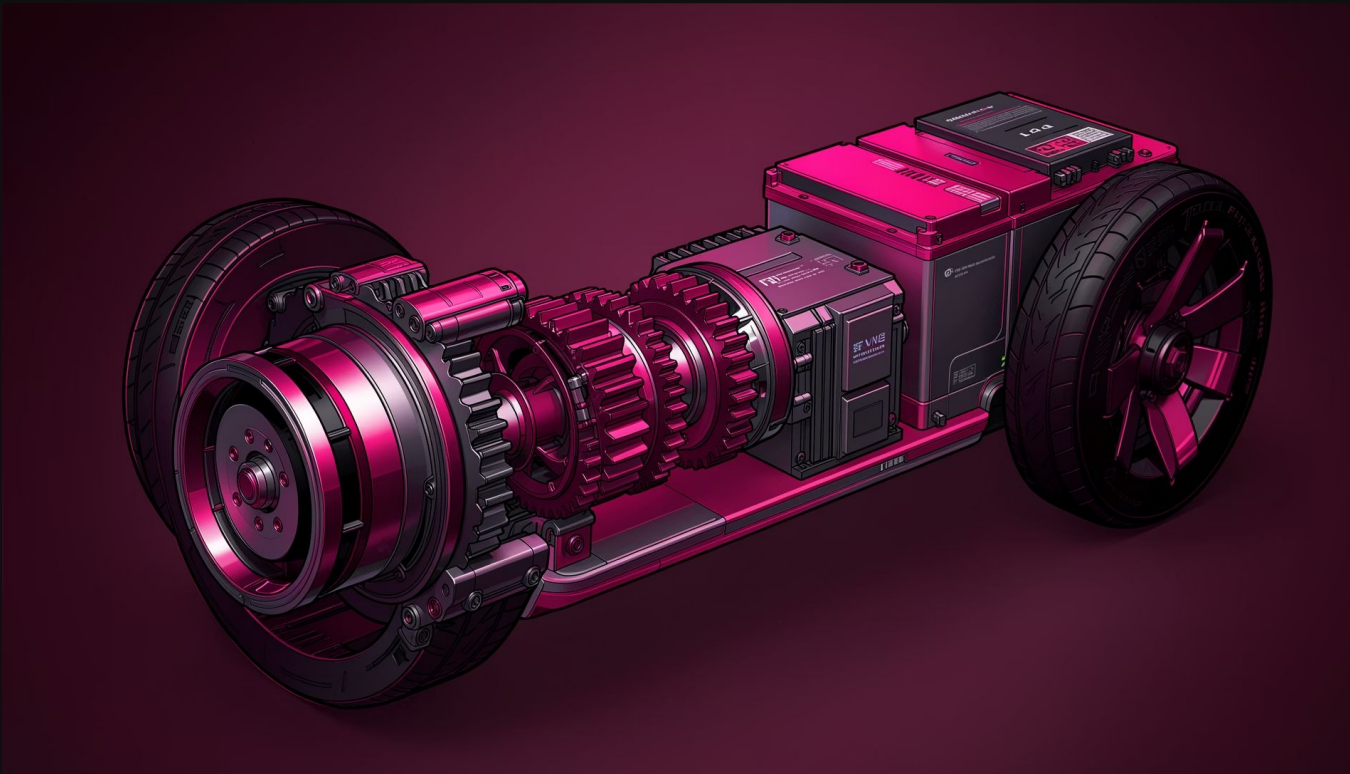
How much of a predictive controller's benefit comes from the forecast itself? Answering it needs a vehicle that can physically follow the test profiles, standard test cycles, forecasts that bracket the learned one, and energy compared at equal tracking quality.



Evaluation Setup

- 1 Test Profiles**
EPA UDDS (city, 1,369 s), HWFET (highway, 765 s) and US06 (aggressive, 600 s), never used for training or tuning
- 2 Forecasts (2 s ahead)**
Persistence · constant acceleration · LSTM · oracle (perfect V2X preview)
- 3 Controllers**
Bang-bang · PID · PID + deceleration feedforward · MPC (2 s horizon, re-planned every 0.4 s)
- 4 Metrics**
Net battery energy per km · speed-tracking RMSE · RMS jerk

Vehicle and Forecaster



EV Digital Twin

- 2,500 kg EV with a 9:1 reduction gear; motor 400 Nm / 150 kW at 92% efficiency
- Regenerative braking up to 200 Nm / 60 kW at 75%, friction brakes for the rest
- Up to 3.85 m/s^2 acceleration, enough for every cycle including US06's 3.8 m/s^2 peak

LSTM Speed Forecaster

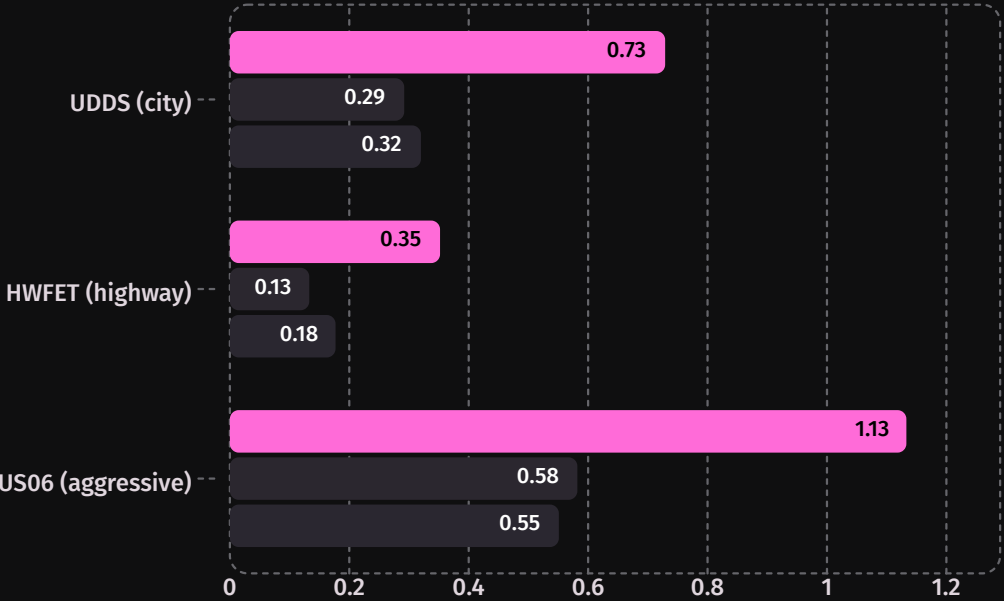
- Two stacked LSTM layers (128 and 64 units), batch normalization, dropout 0.2, dense $32 \rightarrow 20$
- Input: the last 10 s of speed; output: the change in speed over the next 2 s
- Trained on 240 synthetic traffic episodes (urban, highway, aggressive), split by episode; early stopping on validation; 5 seeds

Forecast Accuracy

2 s forecast RMSE in m/s (lower is better). The LSTM halves persistence's error, but a one-line constant-acceleration extrapolation is about as accurate.

Persistence Constant Accel. LSTM

Drive Cycle



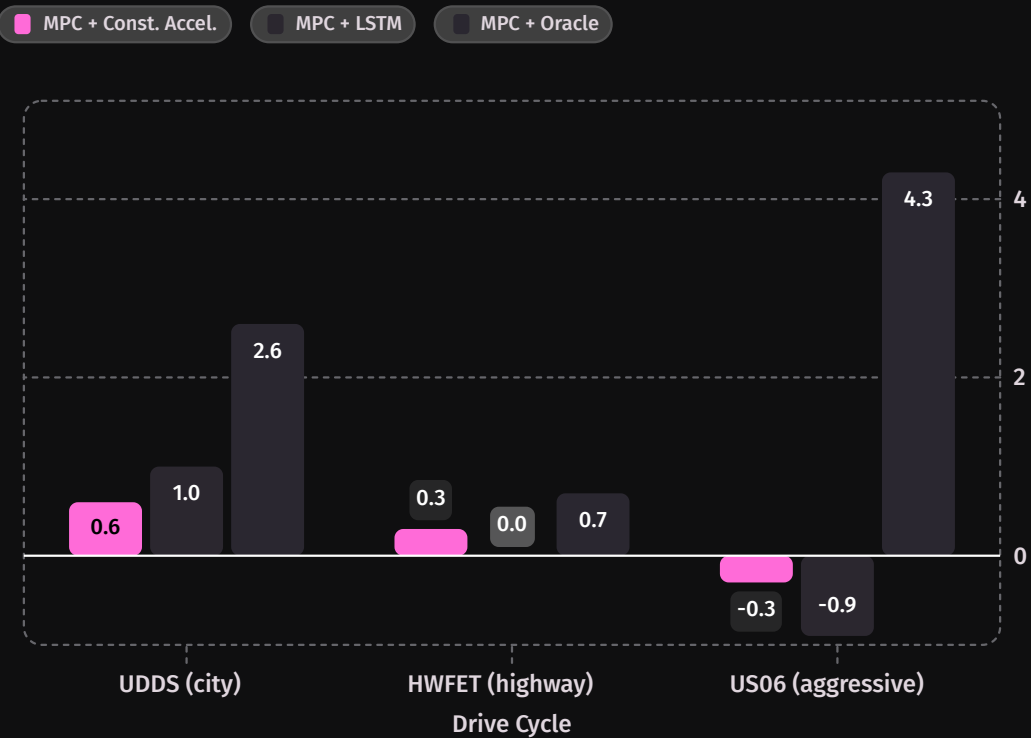
Cycle	Persistence	Const. Accel.	LSTM (5 seeds)
UDDS (city)	0.729	0.292	0.320 ± 0.046
HWFET (highway)	0.352	0.133	0.177 ± 0.045
US06 (aggressive)	1.133	0.582	0.551 ± 0.034

❏ The LSTM halves persistence's error, but a one-line constant-acceleration extrapolation is about as accurate.

Energy at Equal Tracking

Energy saving vs PID at the PID's tracking error:

Cycle	MPC + Const. Accel.	MPC + LSTM (5 seeds)	MPC + Oracle
UDDS (city)	+0.6%	+1.0% ± 0.8	+2.6%
HWFET (highway)	+0.3%	+0.0% ± 0.6	+0.7%
US06 (aggressive)	-0.3%	-0.9% ± 1.7	+4.3%



A perfect preview is worth a few percent, mostly in aggressive driving. The LSTM recovers about 40% of it in the city and none on the highway or the aggressive cycle.

❏ **Key insight:** Oracle MPC yields up to +4.3% savings on US06, but the LSTM MPC delivers -0.9% on the same cycle — highlighting the gap between ideal and learned forecasting.

Why the LSTM Falls Short

Not More Accurate Than Extrapolation

The LSTM is not more accurate than constant-acceleration extrapolation on the city and highway cycles, undermining its core value proposition.

Jittery Forecasts

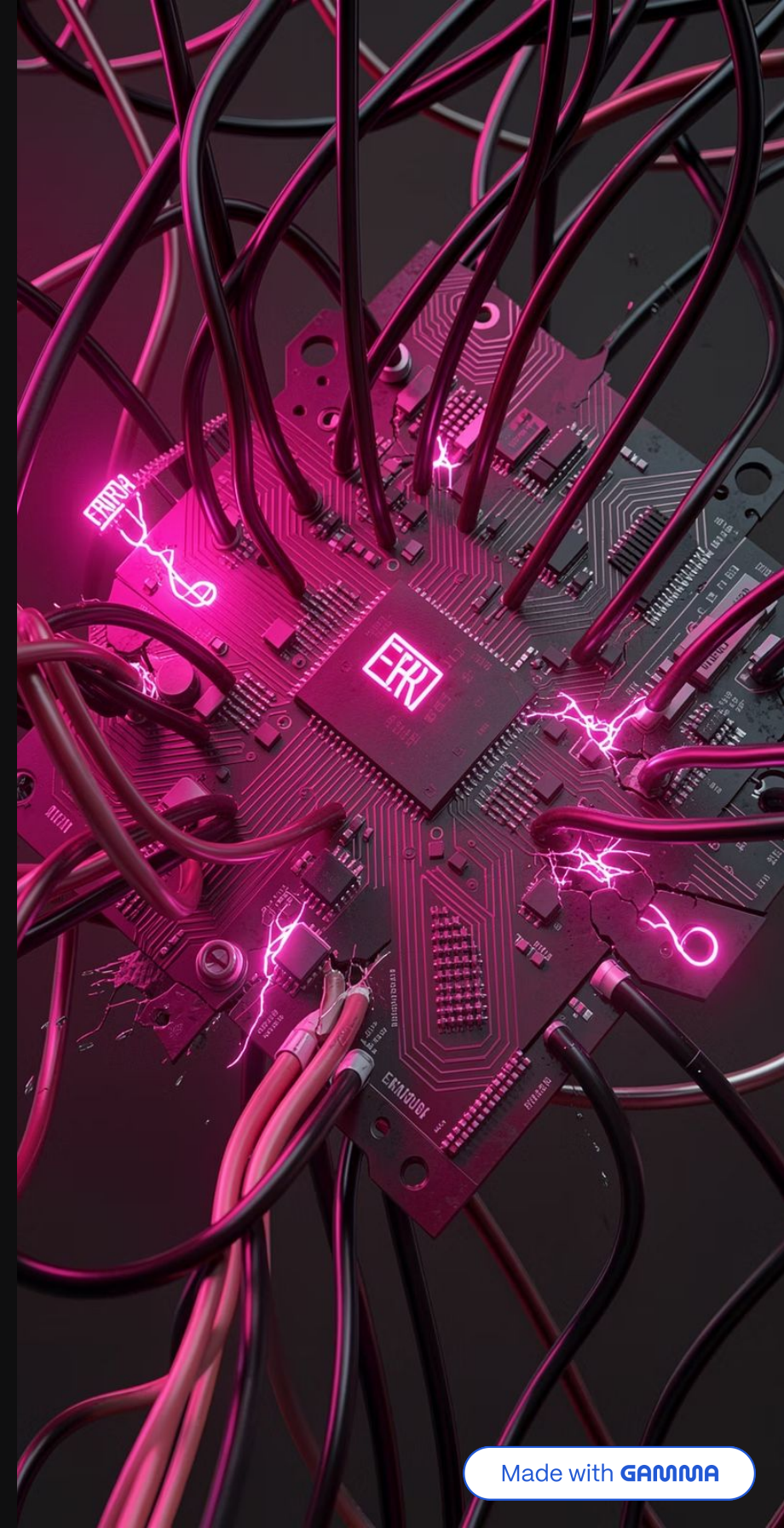
The forecast changes from step to step, the MPC re-plans on each one, and its jerk is 2.5–2.8× the oracle MPC's. The extra acceleration costs energy.

Feedforward Has No Energy Effect

The deceleration-only feedforward changes PID energy by $\leq 0.4\%$, even with a perfect forecast.

Bang-Bang Is Cheap but Rough

1–5% less energy than PID, but 1.5–4.5× worse tracking and 20–80× the jerk — an unacceptable trade-off for passenger comfort.



Conclusions and Future Work

Conclusions

A perfect 2-second preview lets the MPC match the PID's tracking with 0.7–4.3% less energy. A stacked LSTM forecaster recovers little of that, because it is no more accurate than constant-acceleration extrapolation and its forecasts are not smooth.

Next Steps

- Real traffic-speed data from V2X or probe vehicles
- Smoothed forecasts or a smoothness penalty in LSTM training
- Longer V2X preview and road-grade preview
- Safe car-following constraints and multi-vehicle scenarios



References

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